



# Enhancing in-car interaction with On-Device Generative AI

## Overview

This project demonstrates the capabilities of generative AI, utilizing the Llama 3-8B model on the NVIDIA Jetson Orin Nano 8GB, which serves as the benchmark for hardware of the latest in-car head unit chips. The AI was configured to simulate interactions as a Honda vehicle, with future possibility to integrate real-time vehicle data for enhanced interactivity. The model can be optimized or replaced with alternatives to reduce RAM usage, ensuring a lightweight solution. This project highlights the significant potential of generative AI in automotive technology, serving as an essential benchmark as the industry moves forward.

Consider this question:  
In what ways could  
**Generative AI enhance**  
the overall user experience in  
**vehicle interactions?**



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## Key Project Factors

Project Owner: Ryo Fujimura  
Supervisor: Jake Kwon  
Duration: 8 weeks (Summer 2024)  
Hardware: NVIDIA Jetson Orin Nano 8GB  
Software: Linux, JetPack, Llama3-8B



## Presentation Highlights

### Overview of Generative AI

How Generative AI models are made

Data Input → Training Process (Parameter) → Large Language Model

What Generative AI does

Query Input → Large Language Model → Answer Output

AI Model Name Examples: Llama3.1-8B, Llama3.1-70B

Source: NVIDIA, Meta

### Open-Source

Open-Source Software

Linux → Android / Android Automotive / AGX

Open-Source AI Model

Gemma 2, Grok, Mistral AI, Meta

Source: Microsoft, Meta, Linux, Google, Grok, Mistral

### Demo Setup

| Software – Llama3 8B                                                                                                                                                                | Hardware – Jetson Orin Nano                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>LLaMA by Meta</p> <p>Open-Source / Publicly Available*<br/>Natural Language Text Generation<br/>Customizable Model</p> <p>Requires:<br/>30 TOPS<br/>8GB RAM<br/>16GB Storage</p> | <p>NVIDIA JETSON</p> <p>Available for Development<br/>NVIDIA GPU<br/>Access to JetPack SDK</p> <p>Hardware:<br/>40 TOPS<br/>8GB RAM<br/>Optional SSD Storage</p> |

Source: Meta, NVIDIA. \*Requires a license if >700M monthly active users

### What's Next for Honda?

Natural communication

Voice User Preference

Action

Honda Specific Knowledge

App Integration

The presentation starts with an overview of generative AI, ensuring the audience had a shared understanding of the concept.

Open-source platforms like Android and AGX drive automotive innovation. Honda must adopt AI open-source to remain competitive.

Deploying the Llama 3 8B model on Jetson Orin Nano with 8GB RAM, serving as a benchmark for the latest Qualcomm Automotive Chip.

Honda's next step with generative AI includes natural voice, Honda-specific knowledge, action calling, and app integration and more.